Image Processing

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Final Project

Submission Instructions

Before beginning, the Team Leader must submit the names of all teammates followed by the chosen project code (e.g., P1, P2) for approval.

2-Person Projects (Image-Based)

These projects focus on static image classification, computer vision, and machine learning techniques.

- ***P1: Athletic Form Correctness Evaluator***
  - *Task: Analyze a photo dataset of a specific sport or exercise using a machine learning model to determine if the athlete is performing the action with correct form.*
- ***P2: Court Surface Condition Classifier***
  - *Task: Classify the condition of a playing surface (Dry, Wet, Worn, or Damaged) using a close-up image dataset of grass, clay, or hardcourts.*
- ***P3: Static Football Formation Detector***
  - *Task: Use machine learning and image processing techniques to detect a football team's formation based solely on the initial image at kickoff.*
- ***P4: Multi-Sport Image Classifier***
  - *Task: Train and run a classification model on a diverse photo dataset covering a wide variety of sports.*
- ***P5: Static Sprint Start Analyzer***
  - *Task: Analyze a side-view image of an athlete in the sprint blocks. Measure key metrics (shin angle, hip height, back flatness) and score their position against optimal athletic standards.*

- ***P6: Aerial Object Detection System for Human, Ship, and Vehicle Recognition***
  - *Develop a computer vision model capable of detecting and localizing one of the three classes of objects – human, ship, and vehicle – from aerial (satellite or drone) imagery.*

3-Person Projects (Video-Based)

These projects involve dynamic video processing, object tracking, and spatial mapping, requiring a more advanced pipeline.

- ***P1: Dynamic Football Formation Tracker***
  - *Task: Detect and track a football team's tactical formation continuously throughout a live match video dataset.*
- ***P2: Video Sprint Start Analyzer***
  - *Task: Process video footage of a sprint start to dynamically measure changing joint angles (shin, hip, back flatness) and evaluate the launch mechanics against elite standards.*
- ***P3: Dynamic Form & Movement Evaluator***
  - *Task: Build an ML model using a video dataset of a specific sport or exercise to assess execution quality and detect technical errors over time.*
- ***P4: Multi-Sport Ball Tracker***
  - *Task: Implement an object tracking algorithm to accurately follow a ball (football, tennis ball, basketball, etc.) across consecutive video frames.*
- ***P5: Player Tracking & Tactical Heatmap Generator***
  - *Task: Detect players from broadcast or drone footage, map their real-time coordinates onto a 2D pitch model, and generate cumulative team or individual occupancy heatmaps over a match clip.*
- ***P6: Perception System for Autonomous Driving – Real-time Object Detection (Car, Pedestrian, Cyclist, Traffic Sign)***
  - *Design and implement a computer vision model capable of detecting critical road objects – car, pedestrian, cyclist, and traffic sign – in real time from onboard camera images.*

AI Policy & Academic Integrity

The use of Large Language Models (LLMs) is permitted as a development aid. However, your final submission must show deep personal understanding. The code, report, and presentation must look entirely original and human-authored. Obvious AI-generated templates, boilerplate text, or unverified code hallucinations will result in a heavy grading penalty.

Required Deliverables

Your final project submission must contain the following components:

1. **Clean, Runnable Code**
    - *Provide fully documented, well-structured source code.*
    - *Include a brief README file with clear environment setup instructions and dependencies so the code can be executed without troubleshooting.*
2. **Demonstration Video**
    - *Record a short video showing the code executing live and displaying the final results.*
    - *Walk through the core parts of the script to prove the pipeline works end-to-end.*
3. **Project Report**
    - *Submit a comprehensive technical report detailing your methodology, dataset preparation, model architecture, performance metrics, and final conclusions.*
4. **Presentation Slides (PowerPoint)**
    - *Design a polished deck summarizing your project's objectives, technical workflow, results, and key takeaways for the final presentation.*